

Computing Tree Maximum Crown Width Using FVS Equations

Converting FIA CCF equations to MCW equations (Working Draft)

David Marshall

June 24, 2026

0.0 Change Log

6/22/2026 – updated section 5.0 on implementation with an Appendix to compare species with multiple different MCW equation based on Aaron’s suggestion.

1.0 Introduction

The Forest Vegetation Simulator (FVS) variants use three methods to compute crown competition factor (CCF):

1. compute CCF directly (Table 4.5.1).
2. compute maximum crown width (MCW) and use that to compute CCF (Table 4.5.1).
3. compute crown widths (CW) and use that to compute CCF (Tables 4.4.1, 4.4.2, 4.4.3).

The calculation of CCF is documented in section 4.5 of the FVS variant guides and the methods for each variant is summarized in the table below.

Table 1: FVS variants.

Variants	Name
AK	Alaska
BH	Blue Mountains
CA	Inland California and Southern Cascades
CI	Central Idaho
CR	Central Rockies
CS	Central States
EC	East Cascades
EM	Eastern Montana
IE	Inland Empire
KT	Kootenai, Kaniksu, and Tally Lake
LS	Lake States
NC	Klamath Mountains
NE	Northeast
OC	ORGANON SWO
OP	ORGANON Pacific NW
PN	Pacific Northwest
SN	Southern

Variants	CCFtable	CCFnoEQ	CCFDBHbrk	CWtable
PN	CCF	2	1	CCF
SN	MCW	1	0	FCW/OCW
SO	CCF	2	0	CW
TT	CCF	1	1	CW
UT	CCF	1	10/1	CW
WC	CCF	1	10/1	CW
WS	CCF	2	1	CW

The variant map above does not include three variants that overlap other variants: KT (is the northwestern corner of Montana and is included in the IE variant), OP (is ORGANON SMC/NWO variant and is similar to the PN variant), and OS (is the SW Oregon version of ORGANON and is in SW Oregon and would overlap the northern portions of the NC and CA variants and the southern portion of the WC variant).

The following is an example from the Blue Mountains (BM) variant (FVS Staff 2008) section 4.5:

{4.5.1} All species except Pacific yew, Alaska cedar, black cottonwood, and other hardwood

DBH > 1.0": CCFt = R1 + (R2 * DBH) + (R3 * DBH^2)

0.1" < DBH < 1.0": CCFt = R4 * DBH^R5

DBH < 0.1": CCFt = 0.001

{4.5.2} CCF equation for Pacific yew, Alaska cedar, black cottonwood, and other hardwood

DBH > 1.0": CCFt = R1 + (R2 * DBH) + (R3 * DBH^2)

DBH < 1.0": CCFt = (R1 + R2 + R2) * DBH

where:

- CCFt is crown competition factor for an individual tree
- DBH is tree diameter at breast height
- R1 – R5 are species-specific coefficients shown in table 4.5.1

Note that these is a DBH break that is 1.0 inches in this case. In some variants the break is 10 inches (see table 1). The parameters (R1-R5), if used, are give in table 4.5.1 of each FVS variant user's guide. For the BM variant the parameters for Douglas-fir (DF equation 4.5.1 above) and Alaska yellow-cedar (YC equation 4.5.2 above) are:

Table 3: CCF parameters for DF in the Blue Mountains FVS variant.

species	R1	R2	R3	R4	R5
DF	0.0388	0.0269	0.00466	0.0173	1.557
YC	0.0194	0.0142	0.00261	0	0

Note from this example there are three equations used in the various variants, species and DBH combinations.

$$CCFt = R1 + R2 * DBH + R3 * DBH^2 \quad (1a)$$

$$CCFt = R4 * DBH^{R5} \quad (1b)$$

$$CCFt = (R1 + R2 + R3) * DBH \quad (1c)$$

Computing CCF directly (Wykoff and other 1982) adds some computational efficiency but it is problematic if there is a need for the MCW (e.g. to compute largest crown width of forest growth trees, LCW). This note outlines a methodology to recover the MCW values (or equations) from the direct calculation of CCF in FVS.

2.0 Calculation of CCF

The CCF for a stand is computed (Krajicek and other 1961) by summing the maximum crown widths (MCW) multiplied by the per unit are expansion factor for each tree in the tree list.

$$MCW = A + B * DBH$$

$$MCA = pi * (MCW^2) / (4 * (43560/100)) = (pi / (4 * (43560/100))) * MCW^2$$

where DBH is the diameter at breast height and MCW is the vertical projection of the crown width of open-grown trees. The CCF is scaled to be 100 when the tree crowns are just touching (theoretically) with higher values of CCF indicating higher levels of density.

For an individual tree, CCFt = MCA (ignoring the expansion factor). Summing all individual tree CCFt values multiplied by the tree's per unit are expansion factor give the stand CCF. FVS code differentiates the tree CCF from the stand CCF by naming the tree value as CCFt.

Vezina (1963 and referenced by Alexander 1971) presents the calculation of CCFt directly by

$$MCA = pi * (MCW^2) / (4 * (43560/100)) = 0.00180303 * MCW^2$$

$$MCW^2 = (A + B * DBH)^2 = A^2 + 2 * A * B * MCW + (B * MCW)^2$$

$$CCFt = 0.00180303 * (B0 + B1 * DBH + B2 * DBH^2 = CCF)$$

Alexander (1971) gives this derivation for Engelmann Spruce in the Central Rocky Mountains and might have influenced the use of the CCF equation above in FVS (equation 1a from Wykoff and others 1982).

3.0 Calculate MCW from FVS's CCF equations

Below we will demonstrate how to compute MCW from the three CCF equations used in the FVS variants. All these equation will be for individual trees.

3.1 Calculate MCW from CCF equation 1a

The primary CCF equation for FVS variants that compute CCF directly is (typically identified as equation 4.5.1). This is typically for trees with DBH of 1.0-inch and larger, but some variants use 10-inches.

$$CCF = 0.001803026 * MCA = R1 + R2 * DBH + R3 * DBH^2$$

$$MCA = 0.001803026 * (R1 + R2 * DBH + R3 * DBH^2)$$

$$MCW = (MCA / 0.001803026)^{0.5}$$

To demonstrate we will use a Douglas-fir with a DBH of 12-inches and the coefficients for from the BM variant (Table 2).

$$\begin{aligned}
CCF &= 0.0388 + 0.0269 * DBH + 0.00466 * DBH^2 = 1.03264 \\
MCA &= CCF \\
MCW &= (MCA/0.001803026)^{0.5} \\
MCW &= ((0.0388 + 0.0269 * DBH + 0.00466 * DBH^2)/0.001803026)^{0.5} = 23.9317 \text{ feet} \\
CCF_{mcw} &= 0.001803026 * 23.9317^2 = 1.03264
\end{aligned}$$

If we want the parameters for the MCW equation it needs a big more work. Here we will assume that the CCF equations were derived from simple linear MCW equations which are typical.

$$\begin{aligned}
CW^2 &= MCA/0.001803026 = (R1 + R2 * DBH + R3 * DBH^2)/0.001803026 \\
\text{Assuming that the CCF function was derived from a linear CW equation} \\
CW &= B0 + B1 * DBH \\
(B0 + B1 * DBH)^2 &= B0^2 + 2 * B0 * B1 * DBH + B1^2 * DBH^2 \\
\text{which can be rewritten as} \\
(B0 + B1 * DBH)^2 &= (R1 + R2 * DBH + R3 * DBH^2)/0.001803026 \\
\text{So that} \\
B0 &= \sqrt{R1/0.001803026} \\
B1 &= \sqrt{R3/0.001803026}
\end{aligned}$$

Using the BM variant example above and a DBH of 12-inches

$$B0 = \sqrt{R1/0.001803026} = \sqrt{0.0388/0.001803026} = 4.638898 \quad B1 = \sqrt{R3/0.001803026} = \sqrt{0.00466/0.001803026} = 1.607652$$

The differences in the MCW computed directly from the CCF equation and the converted MCW equation is or 0.0009771267 feet or 0.004082982 percent. The difference in computing the CCF directly and from the converted MCW equation is 0.00008432328 or 0.008165797 percent.

3.2 Calculate MCW from CCF equation 1b

For smaller trees and some species other alternative equations (equation 1b and 1c) are used to compute CCF for trees less than a specified DBH. For the BM variant this is 1.0 inches but for some variants it is 10 inches. Below we derive the MCW equation from the CCF equation 1b.

$$\begin{aligned}
CCF &= 0.001803026 * MCA = R4 * DBH^{R5} \\
MCA &= 0.001803026 * (R4 * DBH^{R5}) \\
MCW &= (MCA/0.001803026)^{0.5}
\end{aligned}$$

To demonstrate we will use a Douglas-fir with a DBH of 0.9-inches and the coefficients for from the BM variant (Table 2).

$$\begin{aligned}
CCF &= 0.017299 * DBH^{1.5571} = 0.828732 \\
MCA &= CCF \\
MCW &= ((0.017299 * DBH^{1.5571})/0.0018) = 2.853547 \text{ feet} \\
CCF_{ncw} &< -0.001803026 * 2.853547^2 = 0.828732
\end{aligned}$$

Recovering the MCW equation from the CCF equation (1b) is as follows.

$$\begin{aligned}
MCW &= ((R4 * DBH^{R5})/0.001803026)^{0.5} \\
MCW &= (R4/0.001803026)^{0.5} * (DBH^{R5*0.5}) \\
A1 &= (R4/0.001803026)^{0.5} = \text{sqrt}(0.017299/0.001803026) = 3.097487 \\
A2 &= R5 * 0.5 = 1.5571 * 0.5 = 0.77855 \\
MCW &= 3.097487 * 0.9^{0.77855} = 2.853547
\end{aligned}$$

Which matches the MCW and CCF values computed directly from the CCF equation (1b).

3.3 Calculate MCW from CCF equation 1c

For smaller trees and some species other alternative equations (equation 1b and 1c) are used to compute CCF for trees less than a specified DBH. For the BM variant this is 1.0 inches but for some variants it is 10 inches. Below we derive the MCW equation from the CCF equation 1c. .

$$\begin{aligned}
CCF &= 0.001803026 * MCA = (R1 + R2 + R3) * DBH \\
Rx &= R1 + R2 + R3 \\
MCA &= 0.001803026 * Rx * DBH \\
MCW &= (0.001803026 * Rx * DBH)^{0.5}
\end{aligned}$$

To demonstrate we will use a Alaska yellow-cedar with a DBH of 0.9-inches and the coefficients for from the BM variant (Table 2).

$$\begin{aligned}
Rx &< -0.0194 + 0.0142 + 0.00261 \\
CCF &= Rx * DBH = 0.032589 \\
MCA &= CCF \\
MCW &= ((Rx * DBH)/0.001803026)^{0.5} = 4.251425 \text{ feet} \\
CCF_{cw} &< -0.001803026 * 4.251425^2 = 0.032589
\end{aligned}$$

Recovering the MCW equation from the CCF equation (1c) is given below.

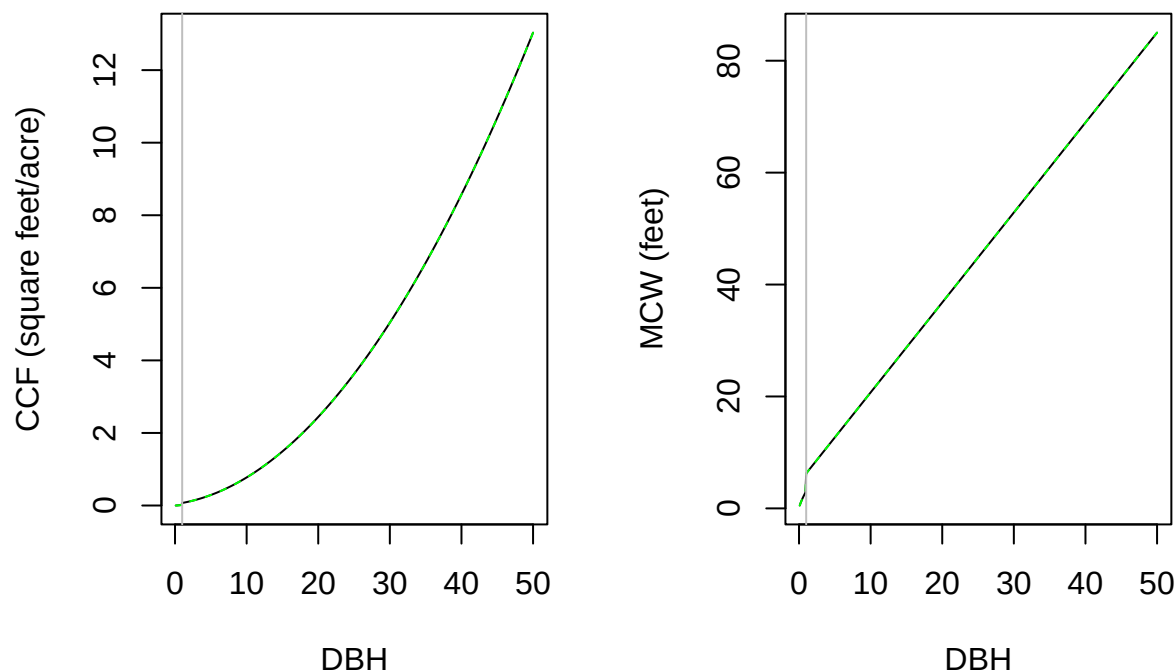
$$\begin{aligned}
MCW &= ((Rx * DBH)/0.001803026)^{0.5} \\
MCW &= (Rx/0.001803026)^{0.5} * DBH^{0.5} \\
C1 &= (Rx/0.001803026)^{0.5} = 4.481395 \\
C2 &= 0.5 \\
MCW &= 4.481395 * DBH^{0.5} = 4.251425
\end{aligned}$$

Where the MCW and CCF estimates computed from the recovered MCW equations match the values computed from the CCF equation above.

3.4 Example

This example demonstrate uses the Blue Mountain FVS variant CCF equations for Douglas-fir to compute the CCF and MCW directly using the equations from CCF equations in section 4.5 (FVS Staff 2008) and

drown in the black lines. These are compared to the MCW and CCF computed using MCW equations (green lines) derived from the CCF equation (4.5.1 for trees 1-inch and larger and 4.5.2 for trees less than 1-inch).



4.0 Crown Width Equation (User Guides section 4.4)

Section 4.4 in the FVS variant user's guides give crown width equations. The crown widths for all variants are defined as maximum crown widths except for CS, LS, NE and SN which all have two tables defined as "Forest-Grown" and "Open-Growth" crown widths (see Table 2). These equations come from a a number of sources.

Note that the CCF calculations for variants CA, CS, LS and NE use the crown with equations of section 4.4 which all the others use the CCF equations of section 4.5.

For the Blue Mountain variant many of the species use an equation fit by Crookston (2005). The only source is a PPT presentation that says the data came from Oregon and Washington CVS inventory plots. This equation predicts crown widths from DBH, HT, CL, BA, EL, BF (forest codes). Given the data source and the predictor variable it seems like this is not MCW of open-grown trees. Using the default BL and EL for the BM variant, $CR = 0.9$ (to compute CL) and a $BA = 0$ appears to give really large CWs. This probably needs more attention of used.

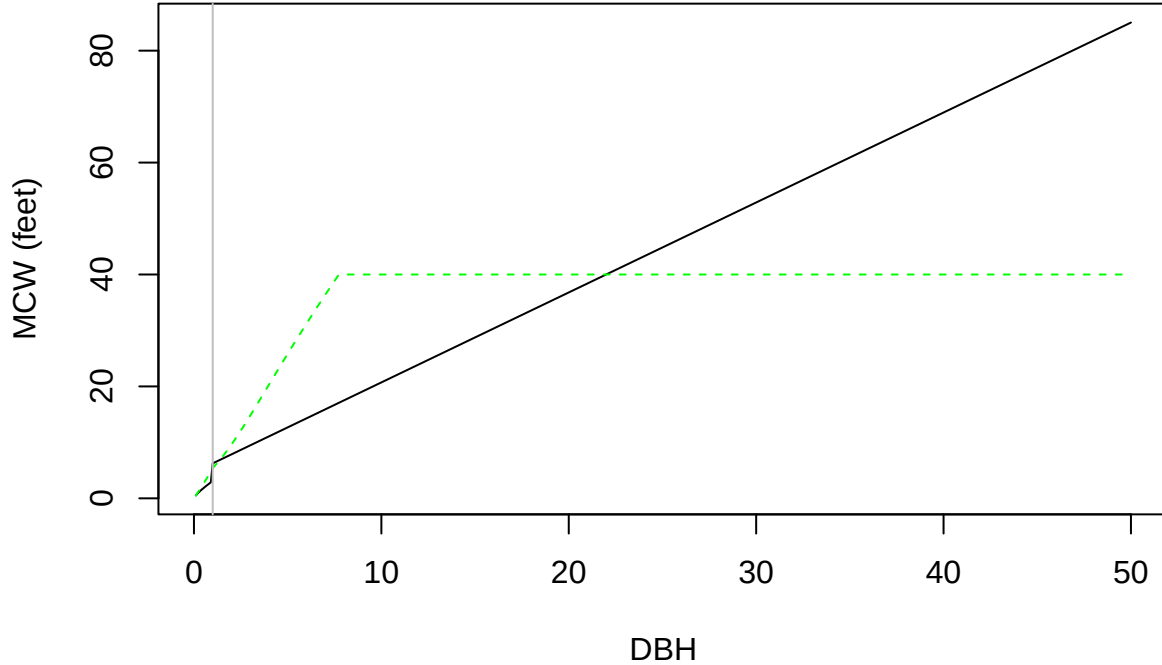
To use these equations the tree-list would have to be filled in with any missing HTs and CRs and the stand BA computed. This is probably not a big issues. The challenge here would be the EL and BF parameters and how to use these (or what impact do they have is a default is used?).

The example below compute the Douglas-fir crown width (MCW) using the Blue Mountain FVS variant coefficients derived from the CCF equation above (black line) and using CW equations (section 4.4) equation 4.4.5 (tables 4.4.1 and 4.4.2) in the BM variant User's Guide.

DBH > MinD: $CW = (a1 * BF) * DBH^2 * HT^3 * CL^4 * (BA + 1.0)^5 * \exp(EL)^6$

DBH < MinD: $CW = [(a1 * BF) * MinD^2 * HT^3 * CL^4 * (BA + 1.0)^5 * \exp(EL)^6] * (DBH/MinD)$

The HT was computed using the variant HT-DBH curves Wycoff logistic type for DF), CL was computed assuming an open-growth CR of 0.95, BA (plot basal area per acre) was set to 0 (open-growth) and the default EL (4500 feet) and BL (1.0) were used. Table 4.4.2 sets a MinD = 1 and puts a 40-foot cap on CW.



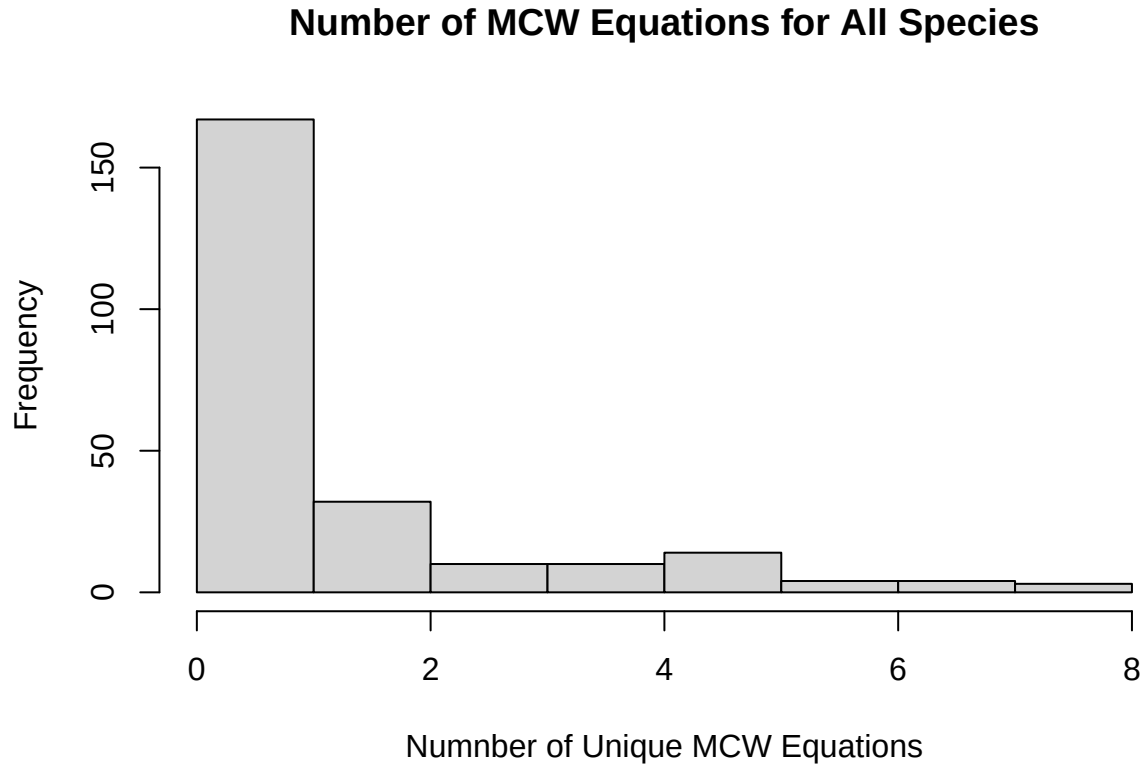
5.0 Implementation Test

An MCW equation data set was created with all the MCW equations for all species and FVS variants (Table 1). The MCW equations were derived from transformed CCF direct calculation equations (section 4.5), MCW equations (section 4.5) and crown width equations (section 4.4) of the variant user's guides. In addition, two separate "variants" were added that contained the published MCW equations of Paine and Hann (1982) and Russell and Weiskittel (2010) that are identified as Variants PH and RW respectively.

Many of the equations have two or three equations representing DBH breaks. The smallest break is usually for DBH values less than 0.1-inch. These equations have not been implemented in this test version which are considering trees with a DBH only.

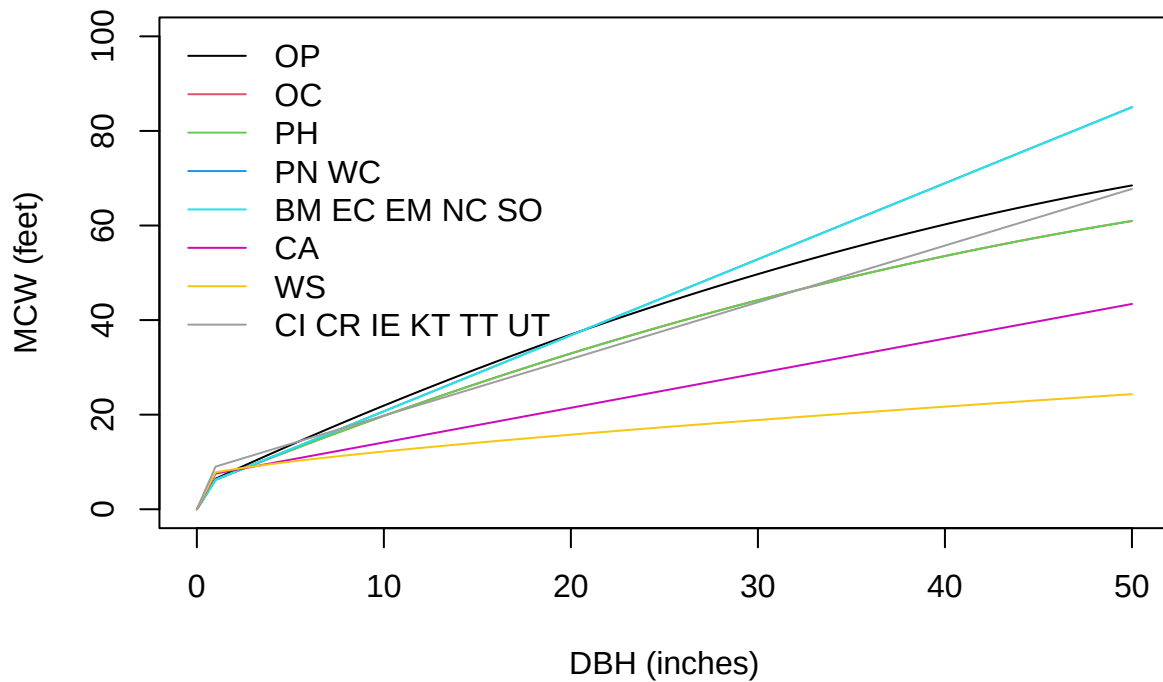
There are a few species that appear to use what I would consider "Forest-Growth" crown widths that reference the work of Bechtold (2003 and 2005) who used inventory data to develop crown width equations (it is not clear yet if these have been modified any for open-growth). The Bechtold equation requires crown ratio, elevation, latitude and longitude. This offers a bit of a challenge (one option used here is to use FVS Variant defaults). This is about 104 species and some of these species may have other equations that use DBH only.

In the MCW equation data set there are 875 equations (including AK) for 244 species (AK changes these numbers by about 1). Of these 164 species have only one MCW equation and 77 species have multiple equations. The number of unique equations for a species range up to 8 for different variants.



Below is an example for Douglas-fir which has 18 MCW equations and 6 unique MCW equations which are graphed below. See the Appendix for similar graphs for all species that have more than one MCW equation.

FVS Variant MCW Equations for DF (202)

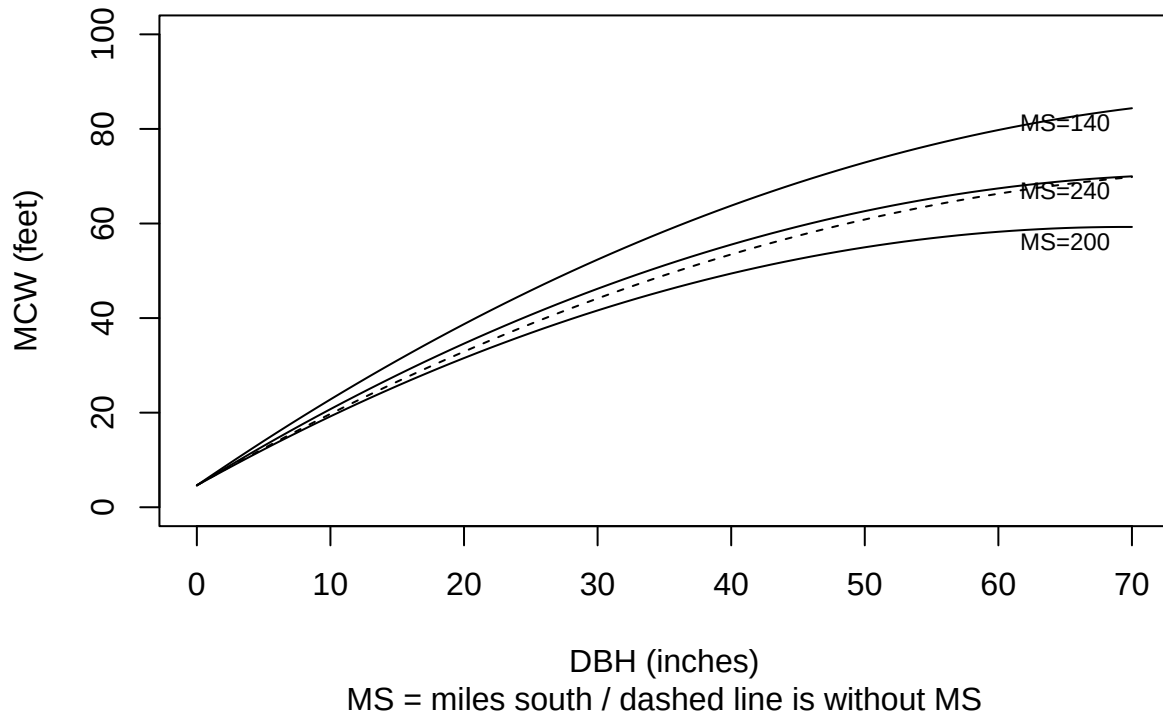


For a CONUS wide MCW application a single equation would have to be selected for species with multiple equations. There are about 471 FIA species in the CONUS/Alaska data set leaving about 227 species (minimally) that would need to be mapped (not counting the > 999 species codes of which there are a few in the CONUS modeling data set). This mapping could be done by genus or softwood/hardwood classifications.

A couple questions this raises:

1. How much difference does adding things other than DBH really make in computing CCF and other variables? Both Bechtold (2003 and 2004) papers give DBH only equations and equations with CR, elevation, latitude and longitude. The Paine and Hann (1982) paper give equations with latitude and longitude and shows a several foot difference (greater or lesser depending on the longitude given as miles south).

Douglas-fir MCW with geographic variables Paine and Hann (1982, Figure 1)



2. Are the differences for equations of the same species really important or how much are they data driven?

For a give stand I suspect these might be of minor significance since we are really just “ranking” species competitive status. However, if one is concerned across a region and that region has differences in MCW for the same DBH how big of an issue is that? For example Paine and Hann (1982) use latitude and longitude and show a 10 foot difference at the extremes of their data range (look at the southern variants CS/WS compared to the more northern variants PN/OP in the graph above).

3. Many (most) of these equations are simple linear with no asymptotic behavior and MCW values getting larger as DBH gets larger. In some cases there are maximum MCW values given as maximum DBH or maximum CW, but most do not have maximums. With the FIA data there are possibly large DBH values and maximums (or some type of constraint) may be needed.

6.0 Discussion

The above shows how to recover the MCW from the CCF equation (section 4.5 of the FVS variant User’s Guides) given for most of the FVS variants (not provided for CA, CS, LS and NE variants which use crown width equations). Most of the MCW equations appear to be typical linear crown width equations although some are power functions depending on variant, species and DBH. Converting the CCF equations to MCW is not difficult (as long as the assumption of a simple linear model is true) but is complicated by the differences in equations (3 equation types) that depend on species and diameter breaks for different variants.

Note from the example plot above that there is a noncontinuous MCW relationship between equation 1 (greater than DBH break) and 2 (less than DBH break) which is 1.0-inch for the BM variant but can be larger, up to 10-inches).

FVS sets CCF to 0.001 for trees smaller than 0.1-inch. For the current project we will not be dealing with trees less than 0.1-inch (4.5-feet) so this should not be a problem. For trees less than 1-inch it might not be a problem to ignore the less than 1.0-inch equation (needs testing). However, in cases that the DBH break is 10-inches this would need to be checked before relying only on equation 1a.

The computed MCW can be used to compute CCF. The crown closure at the tree tip (CCH) is computed using a crown profile and the largest crown width (LCW). Hann (1997) scales MCW to get LCW. Bechtold (2003 and 2004) predicts it (crown width of forest growth trees) directly. The Bechtold (2003 and 2004) equations appear to be implemented in FVS with equations for 87 eastern species and 53 western species. In both papers, Bechtold (2003 and 2004) presents a full model (which include crown ratio, elevation, latitude and longitude) and a reduced equation (includes only DBH).

7.0 References

- Alexander, R.R. 1971. Crown competition factor (CCF) for Engelmann spruce in the Central Rocky Mountains. USDA Forest Service, Rocky Mountain Forest and Range Experiment Station. Research Note RM-188. 4p.
- Bechtold, W.A. 2003. Crown-diameter prediction models for 87 species of stand-grown trees in the eastern United States. *Southern Journal of Applied Forestry*. 27(4): 269-278.
- Bechtold, W.A. 2004. Largest-crown-width prediction models for 53 species in the western United States. *Western Journal of Applied Forestry* 19(4):245-251.
- Crookston, Nicholas L. 2004. Allometric Crown Width Equations for Northwest Trees (PPT) <https://www.slideserve.com/abdalla/allometric-crown-width-equations-for-northwest-trees-powerpoint-ppt-presentation>
- FVS Staff. 2008 (revised September 23, 2025). Blue Mountains (BM) Variant Overview – Forest Vegetation Simulator. Internal Rep. Fort Collins, CO: U. S. Department of Agriculture, Forest Service, Forest Management Service Center. 60p.
- Hann, D.W. 1997. Equations for predicting the largest crown width of stand-grown trees in western Oregon. Research Contribution 17. Forestry Research Laboratory, Oregon State University, Corvallis. 14 p.
- Hann, D.W. 1999. An adjustable predictor of crown profile for stand-grown Douglas-fir trees. *Forest Science* 45:217-225. {ORGANON}
- Krajicek, J.E.; Briinkman, K.A.; Gingrich, S.F. 1961. Crown Competition – A Measure of Stand Density. *Forest Science* 7(1):35-42.
- Russell, M.B.; Weiskittel, A.R. 2010. Estimating tree crown widths for the primary Acadian species in Maine. Joint meeting of the Forest Inventory and Analysis (FIA) symposium and the Southern Mensurationists. Pages 45-52.
- Paine, D.P.; Hann, D.W. 1982. Maximum crown-width equations for southwestern Oregon tree species. Research Paper 46. Forestry Research Laboratory, Oregon State University, Corvallis. 20 p.
- Vezina, P. E. 1963. More about crown competition factor. *Forestry Chronicle* 39: 313-317.
- Wykoff, William R.; Crookston, Nicholas L.; Stage, Albert R. 1982. User's Guide to the stand prognosis model. Gen. Tech. Report INT-133. Ogden, UT: U.S. Department of Agriculture, Intermountain Forest and Range Experiment Station. 112 p.

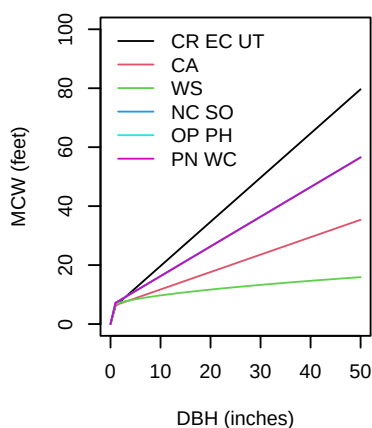
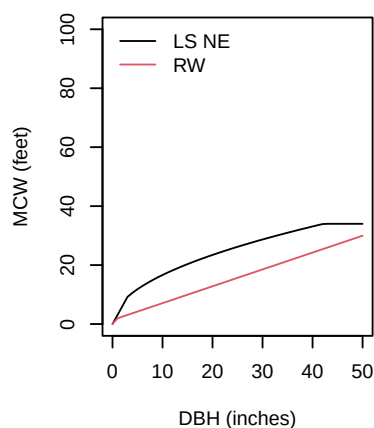
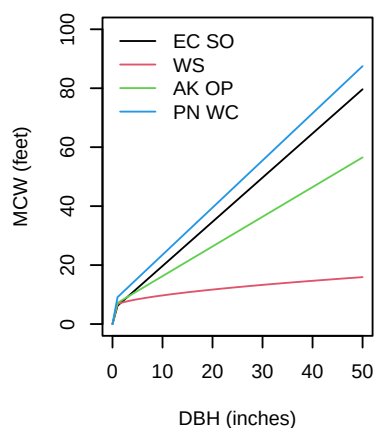
8.0 Appendix – Comparisons of species with multiple MCW equations (SDCD = FIA species code).

The graphs below show the MCW-DBH curves for species that have more than one MCW curves across FVS variants. In some cases the curves may show breaks which are caused by different equations above and below DBH breaks. Some curves (variants) also have either maximum DBH or maximum CW values and these were used and can be seen as horizontal lines.

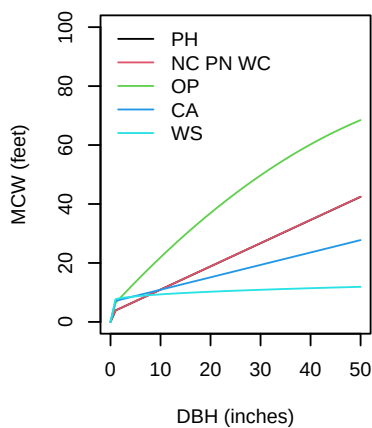
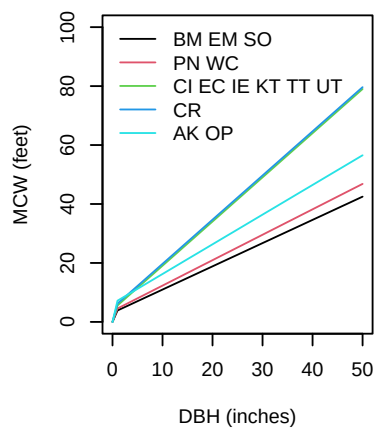
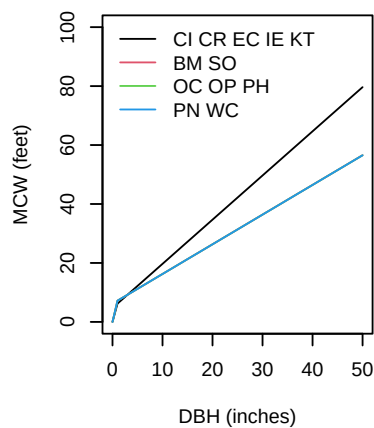
The Bechtold (2003, 2004) crown width models (used in the CS, LS, NE, SN variants) are shown as dashed lines. These assumed a CR = 90 and used the default elevation, latitude and longitude for the FVS variant. All other curves just use DBH as the predictor variable.

The FVS variant definitions are given in Table 1. Also included are curves labeled PH and RW which are not FVS variants but are the Paine and Hann (1982) and Russell and Weiskittel (2010) published MCW curves for comparison.

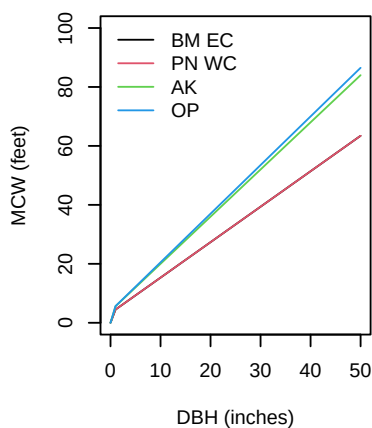
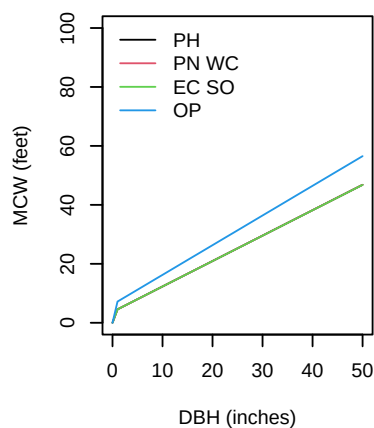
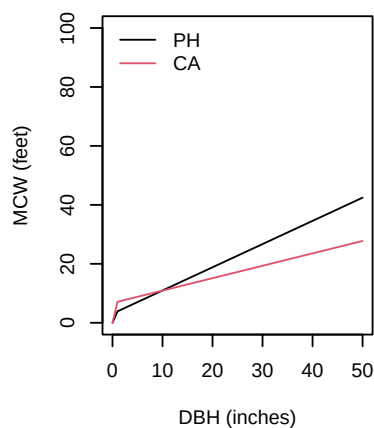
FVS Variant MCW Equations for SF FVS Variant MCW Equations for BFFVS Variant MCW Equations for WF



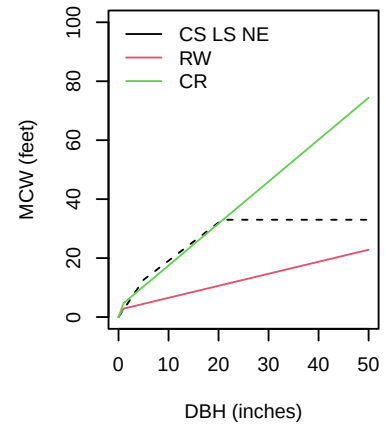
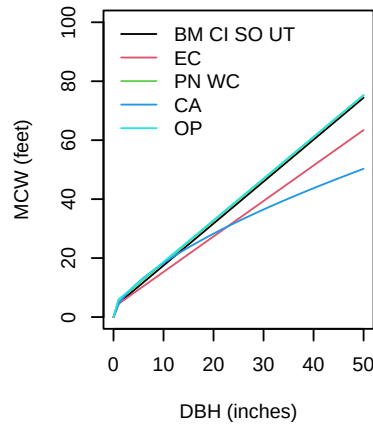
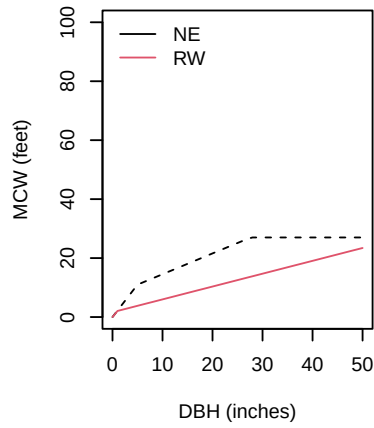
FVS Variant MCW Equations for GFFVS Variant MCW Equations for AFFVS Variant MCW Equations for RF



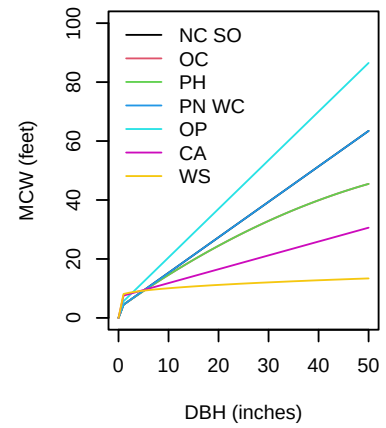
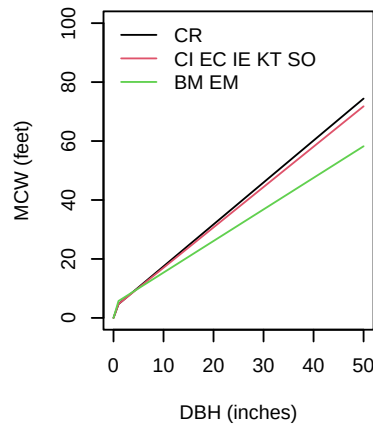
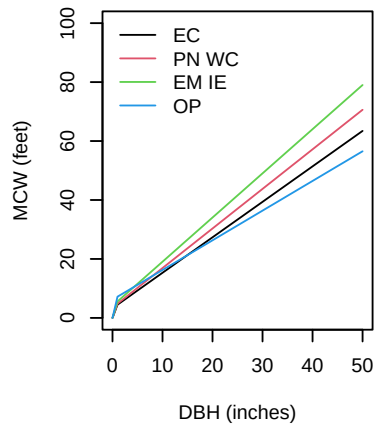
FVS Variant MCW Equations for SHFVS Variant MCW Equations for NFFVS Variant MCW Equations for YC



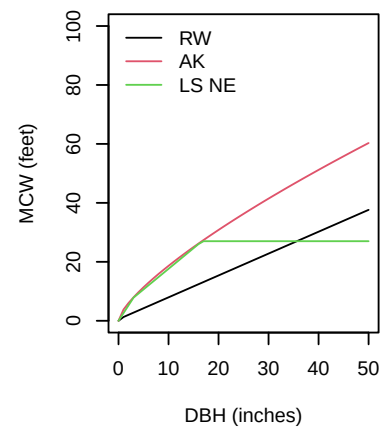
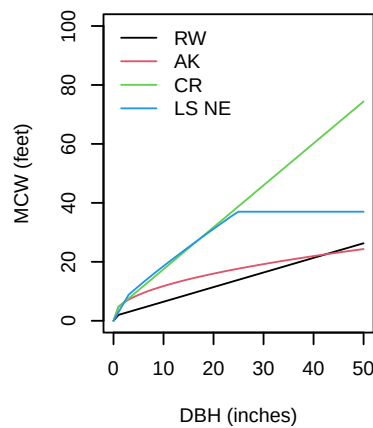
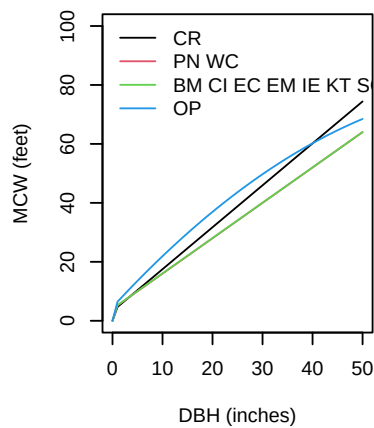
FVS Variant MCW Equations for AWFVS Variant MCW Equations for WJFVS Variant MCW Equations for ER



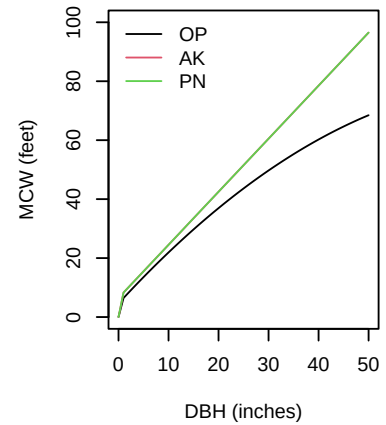
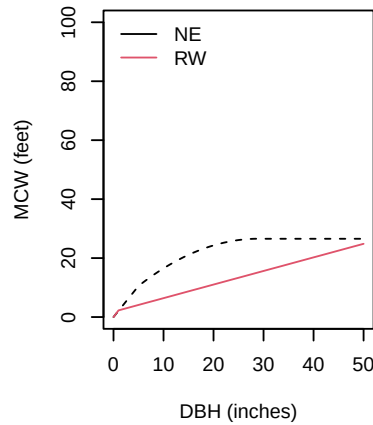
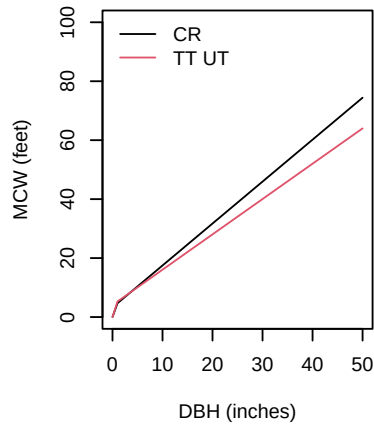
FVS Variant MCW Equations for LL FVS Variant MCW Equations for WL FVS Variant MCW Equations for IC



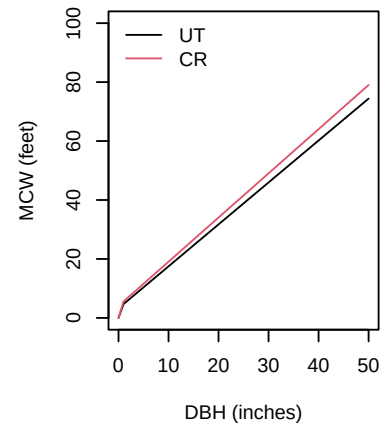
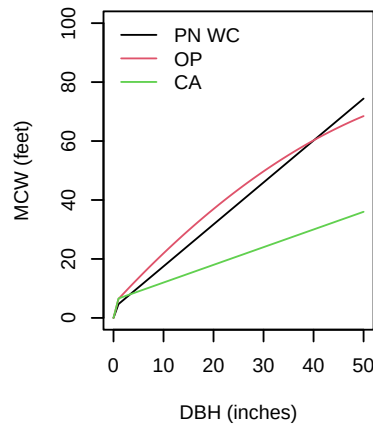
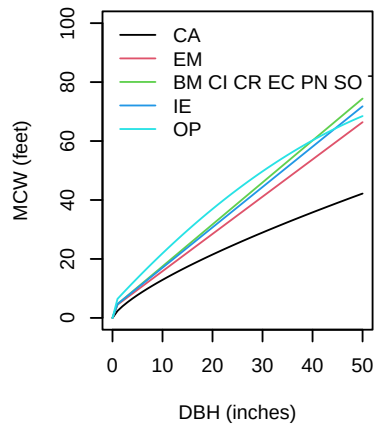
FVS Variant MCW Equations for ESFVS Variant MCW Equations for WSFVS Variant MCW Equations for BE



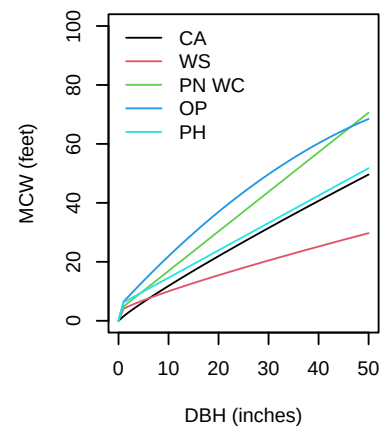
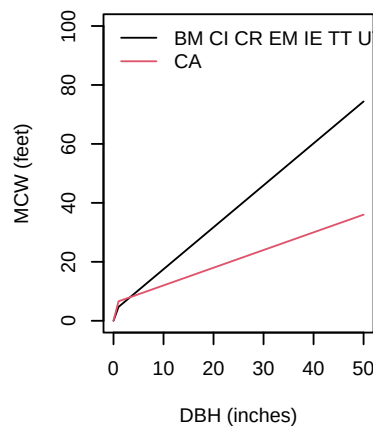
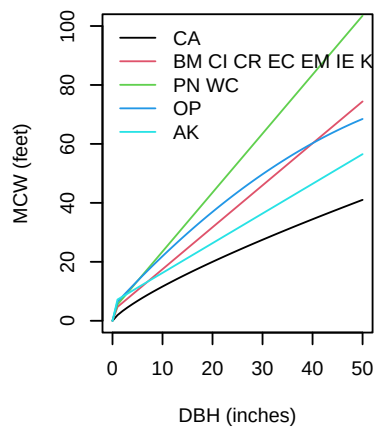
FVS Variant MCW Equations for BS FVS Variant MCW Equations for RS FVS Variant MCW Equations for SS



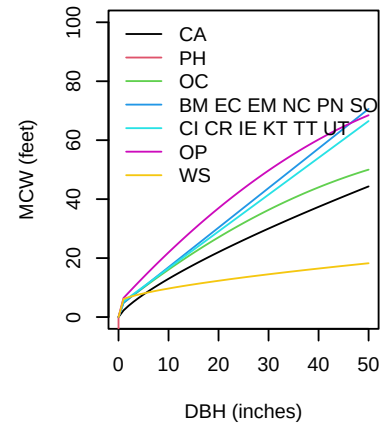
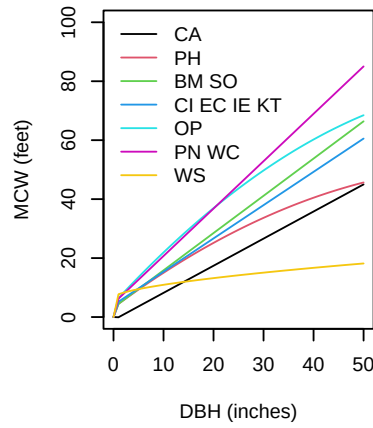
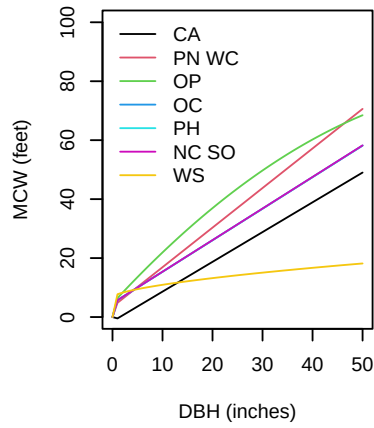
FVS Variant MCW Equations for WB FVS Variant MCW Equations for KP FVS Variant MCW Equations for PI (



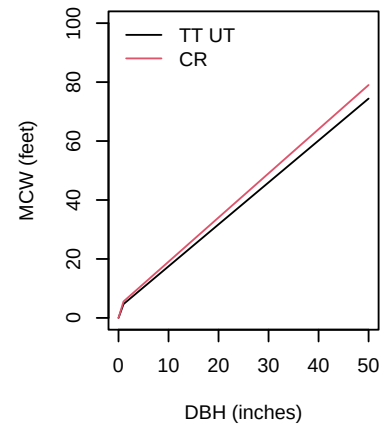
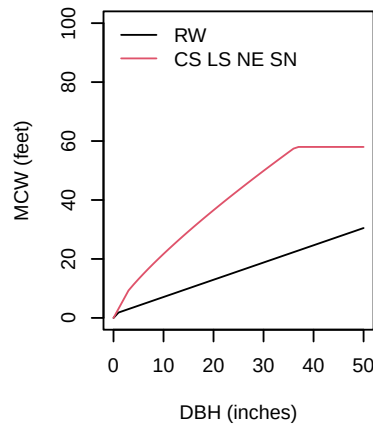
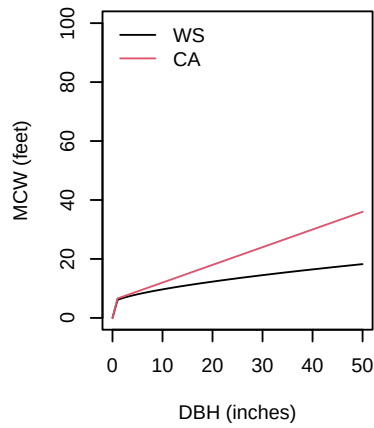
FVS Variant MCW Equations for LP FVS Variant MCW Equations for LM FVS Variant MCW Equations for JP (



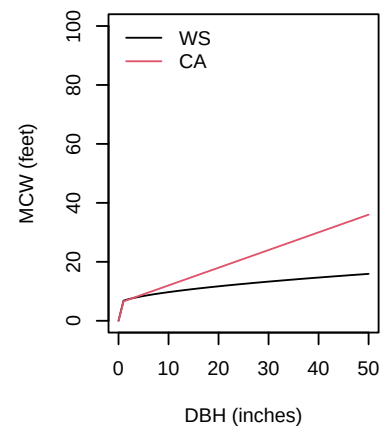
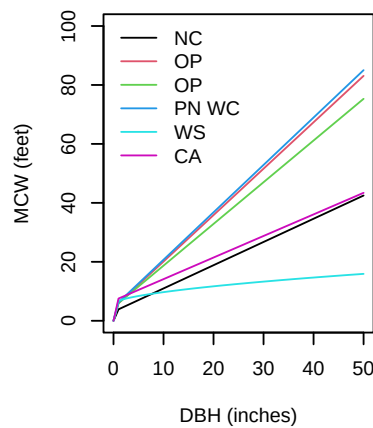
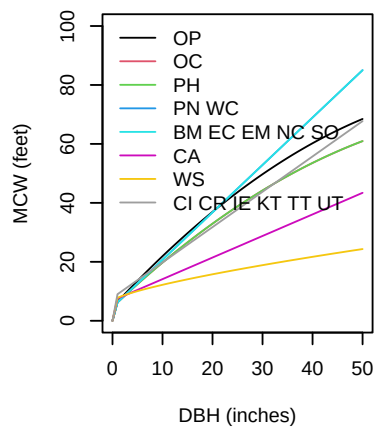
VS Variant MCW Equations for SP VS Variant MCW Equations for WP VS Variant MCW Equations for PP (



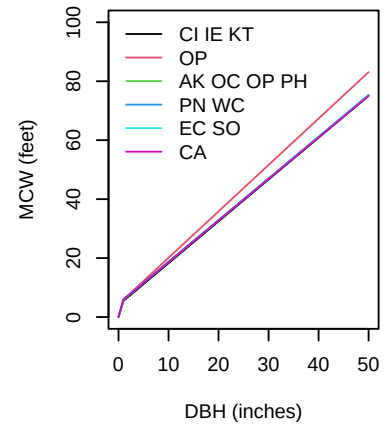
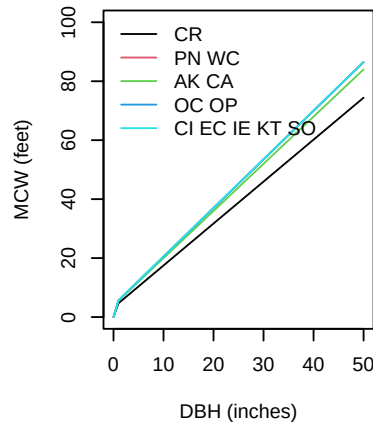
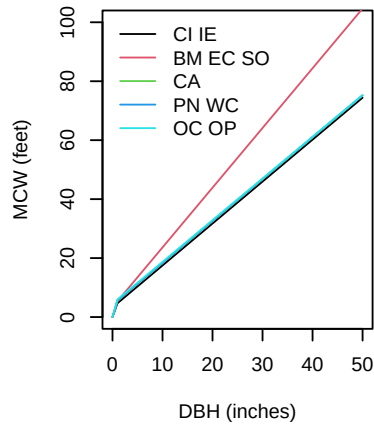
VS Variant MCW Equations for MP VS Variant MCW Equations for WP VS Variant MCW Equations for PM (



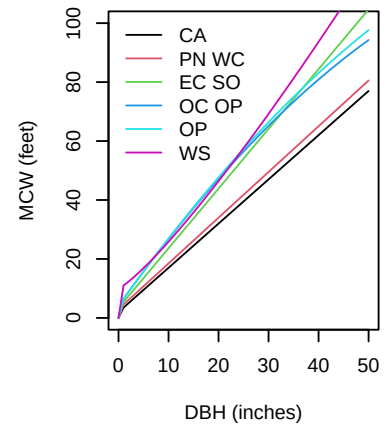
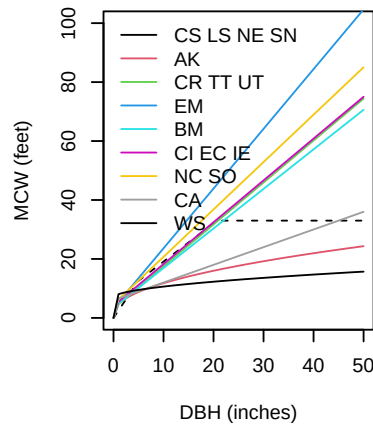
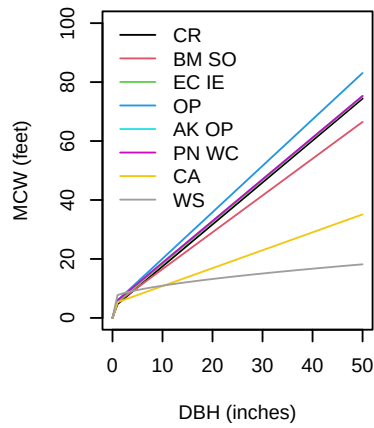
VS Variant MCW Equations for DF VS Variant MCW Equations for RW VS Variant MCW Equations for GS (



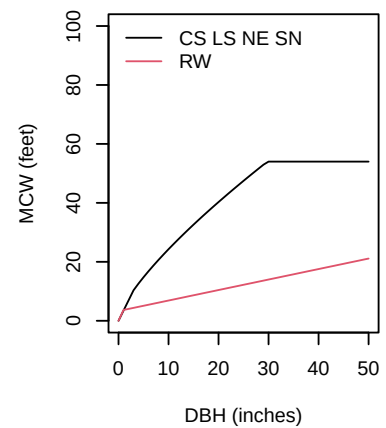
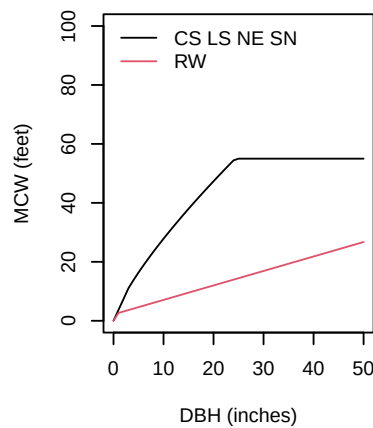
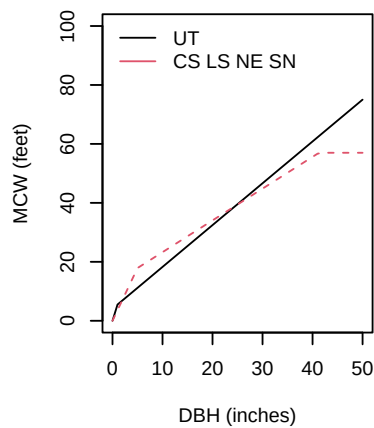
VS Variant MCW Equations for PY VS Variant MCW Equations for RC VS Variant MCW Equations for WH



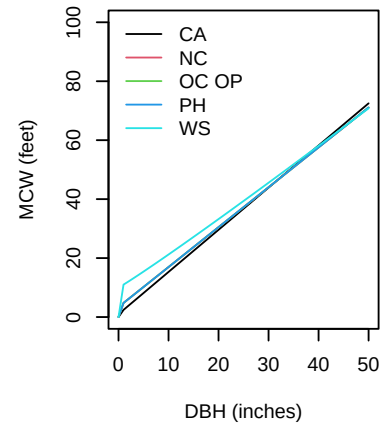
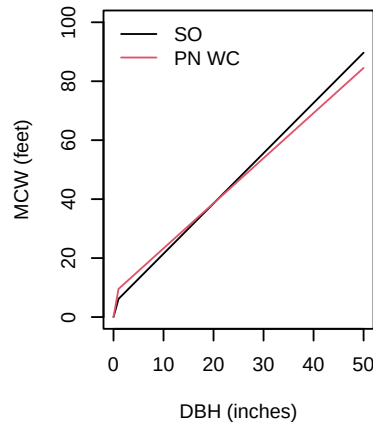
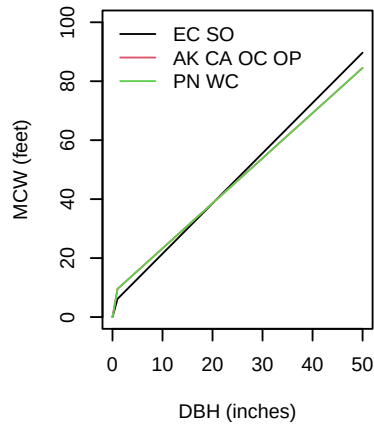
VS Variant MCW Equations for MH VS Variant MCW Equations for OS VS Variant MCW Equations for BM



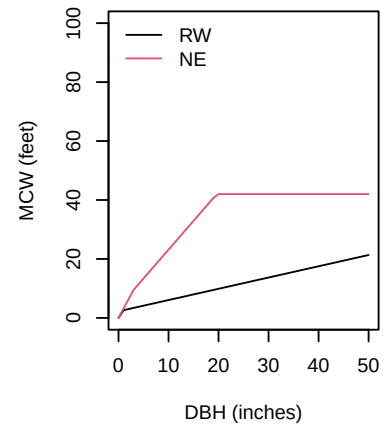
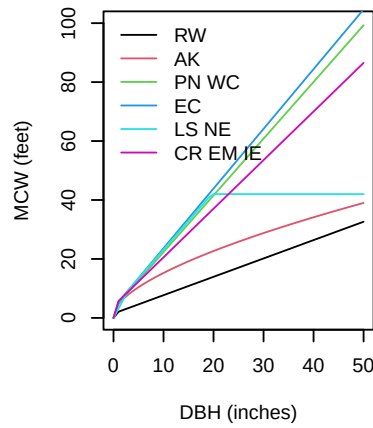
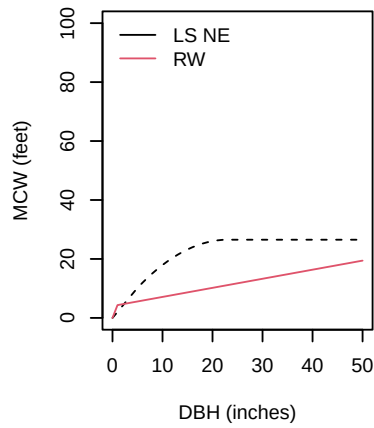
VS Variant MCW Equations for BE VS Variant MCW Equations for RM VS Variant MCW Equations for SM



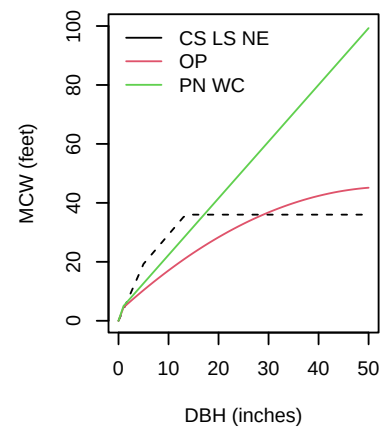
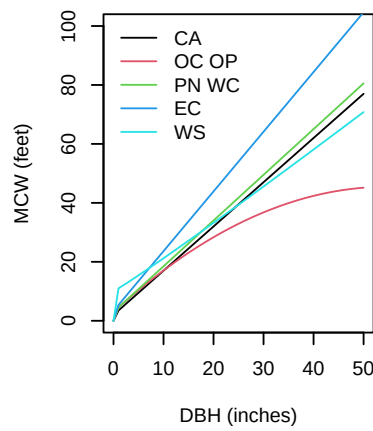
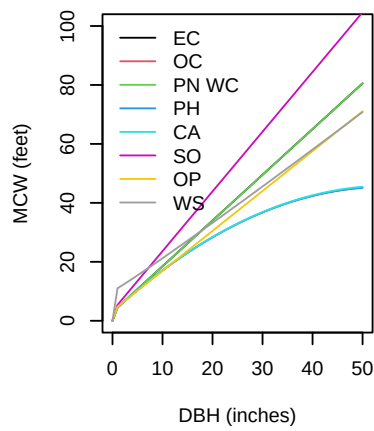
VS Variant MCW Equations for RA VS Variant MCW Equations for WA VS Variant MCW Equations for MA



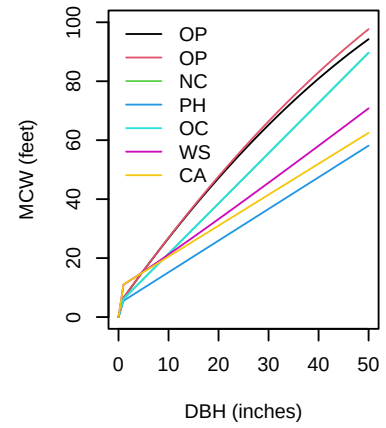
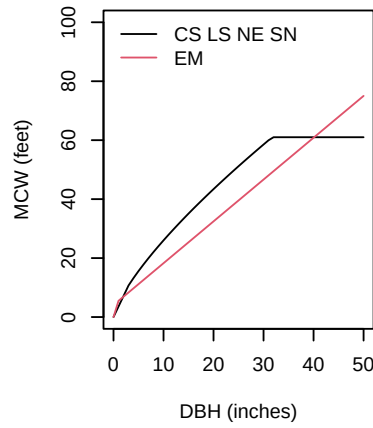
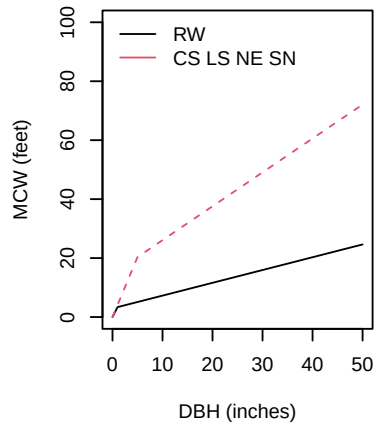
VS Variant MCW Equations for YB VS Variant MCW Equations for PB VS Variant MCW Equations for GB



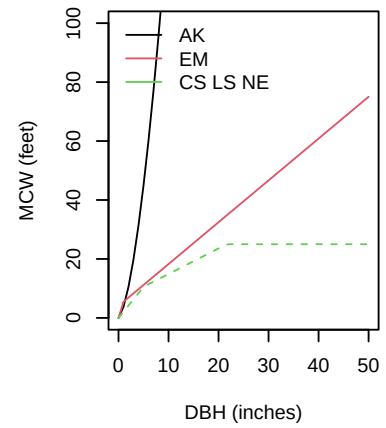
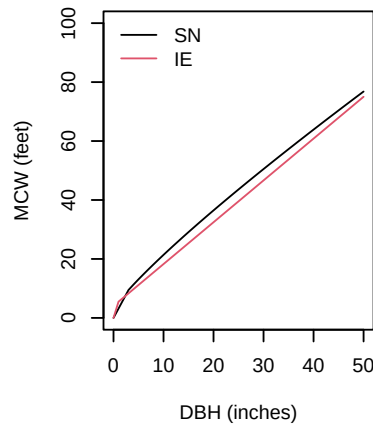
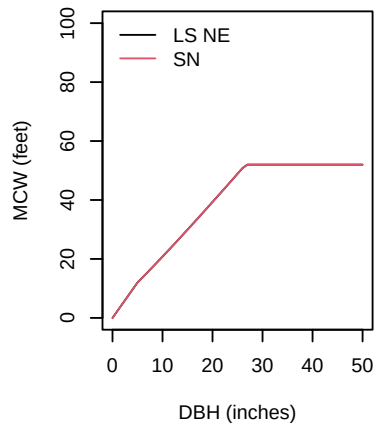
VS Variant MCW Equations for GC VS Variant MCW Equations for DG VS Variant MCW Equations for HT



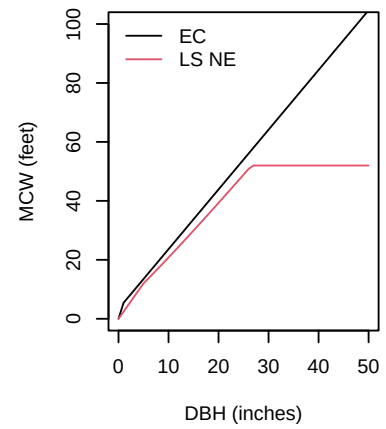
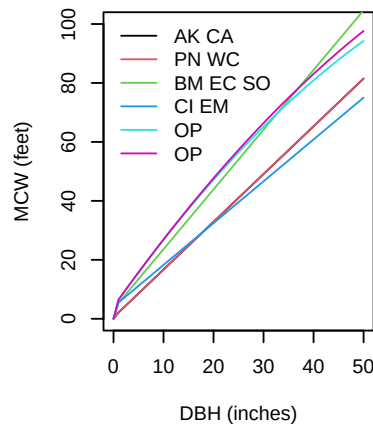
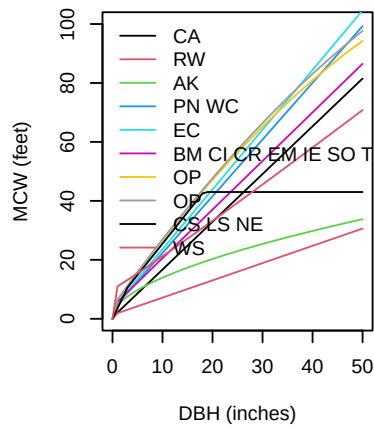
VS Variant MCW Equations for AB VS Variant MCW Equations for GA VS Variant MCW Equations for TO (



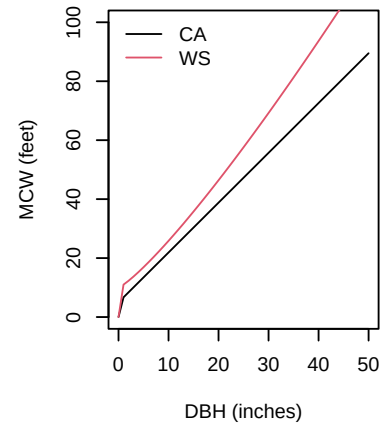
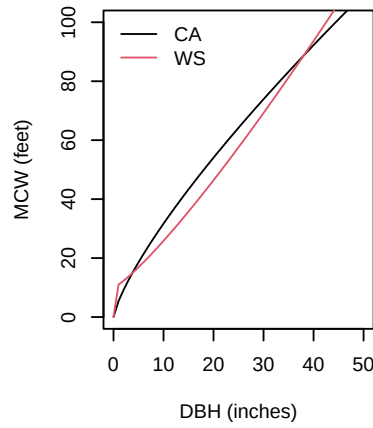
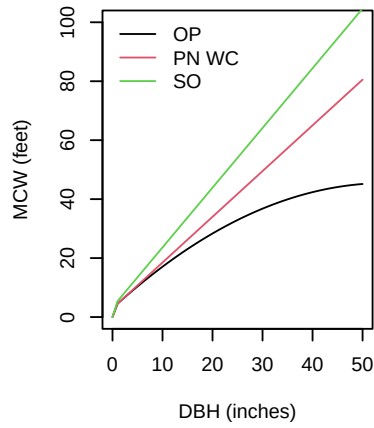
VS Variant MCW Equations for AP VS Variant MCW Equations for CO VS Variant MCW Equations for BA (



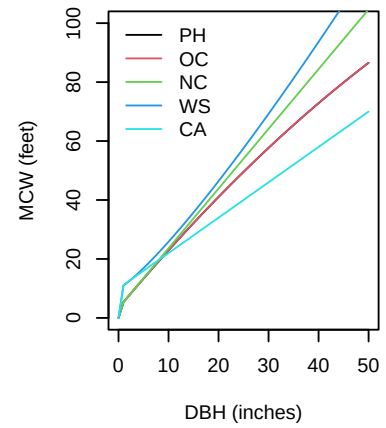
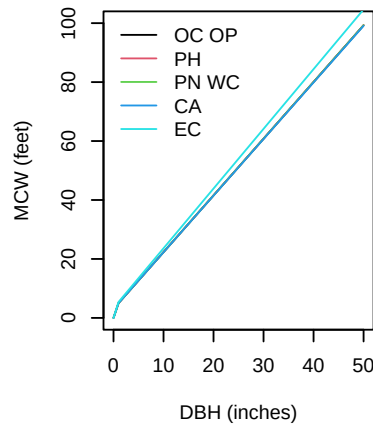
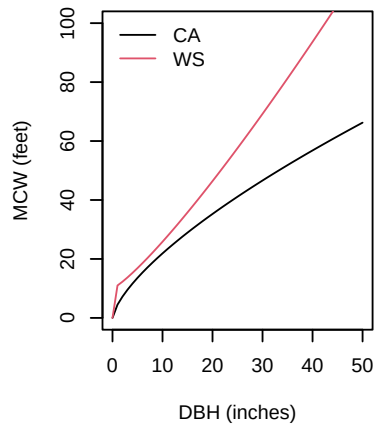
VS Variant MCW Equations for AS VS Variant MCW Equations for CW VS Variant MCW Equations for PL (



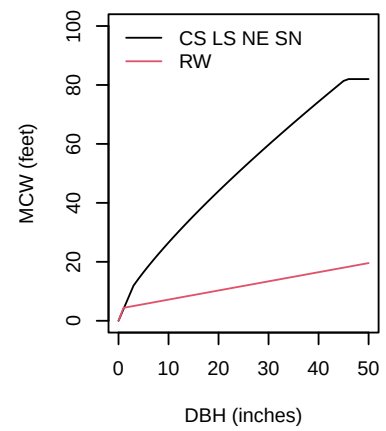
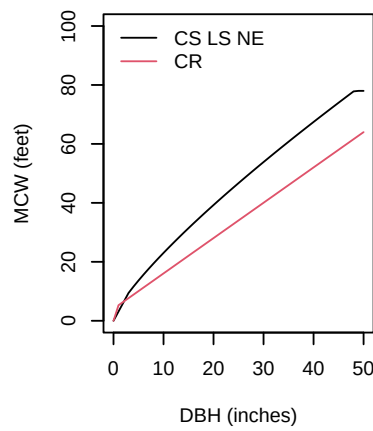
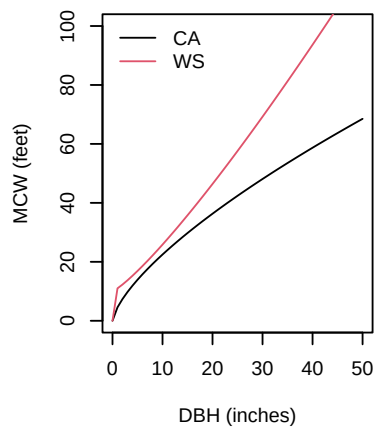
VS Variant MCW Equations for CH VS Variant MCW Equations for LO VS Variant MCW Equations for CY (



VS Variant MCW Equations for BL VS Variant MCW Equations for WO VS Variant MCW Equations for BO (



VS Variant MCW Equations for VO VS Variant MCW Equations for BK VS Variant MCW Equations for RO (



VS Variant MCW Equations for CL VS Variant MCW Equations for OH VS Variant MCW Equations for OT (

